Study Guide AZ-104 — Azure Administrator

> 🌐 **Language:** English | Nederlands

Duration: 8 weeks · **Lab preset:** Minimal (DC01 · W11-01) — Azure tasks run in the cloud **MS Learn path:** Azure Administrator **Exam weight:**

Domain	Weight
Manage Azure identities and governance	20–25%
Implement and manage storage	15–20%
Deploy and manage Azure compute resources	20–25%
Implement and manage virtual networking	15–20%
Monitor and maintain Azure resources	10–15%

> **Updated:** Skills measured as of April 17, 2026. Domain weights and structure unchanged. Minor updates to storage (blob lifecycle, versioning), compute (VM encryption at host, Scale Sets), networking (VNet troubleshooting), and monitoring (Site Recovery failover now explicitly in scope).

> **Prerequisite:** Azure subscription via MSDN/Visual Studio Subscriptions (monthly credit) The SSW-Lab VMs serve as a *hybrid on-premises endpoint* for some labs (VPN, Azure Arc) **Tenant note:** the shared dev tenant helps with Entra and identity topics, but most of AZ-104 still requires real Azure resources. Follow the guide explicitly when it tells you to switch from local VM work to the tenant or Azure subscription.

How to use this study guide

- Start each week by reading the learning objectives and deciding which tasks must happen in Azure itself versus which tasks are only supported by SSW-Lab.
- Work cost-consciously throughout the lab: note what you create and remove resources on the same day when they are no longer needed.
- Complete the knowledge check only after doing both the portal and CLI/PowerShell work; AZ-104 often tests whether you recognise multiple administration paths.
- If you work in a mixed Dutch/English team, keep key terms aligned in both languages, such as *resource group / resourcegroep* and *management group / beheerlaag*.

Lab coverage and expectations

- **Strong SSW-Lab support:** hybrid identity fundamentals, Azure CLI from Windows clients, Azure File Sync integration, networking concepts, and selected administration and monitoring exercises.
- **Partial coverage:** many AZ-104 objectives live primarily in Azure; the local VMs mainly support Arc, VPN, File Sync, and hybrid identity scenarios.
- **Cloud-first areas:** compute, storage, VNets, load balancing, monitoring, backup, and governance should mainly be practised in the Azure portal and via CLI/PowerShell.
- Treat this as a complete study route, but not as a local-only lab: the real exam fit comes from consistently creating and cleaning up Azure resources yourself.

How to use the knowledge checks

- Answer every question by first identifying the correct scope and management layer.
- Verify that you understand not only the portal path, but the underlying platform concept as well.
- Revisit questions on storage redundancy, networking, and RBAC versus Policy, because these are easy to mix up in scenario questions.
- Tie each answer back to operational impact: cost, availability, security, and management overhead.

Week 1 — Azure identities and governance

> **Exam domain:** Manage Azure identities and governance · **Weight:** 20–25%

> **Real-world scenario:** A Sogeti consultant is onboarding a newly acquired subsidiary of a financial services client. The subsidiary has its own Active Directory, 200 employees, and needs scoped access to specific Azure workloads without gaining visibility into the parent company's subscriptions. You are asked to design the RBAC delegation model, create a management group hierarchy, and enforce a policy that restricts resource creation to approved regions only.

Learning Objectives

- [ ] Create and manage Entra ID users and groups, including guest (B2B) accounts
- [ ] Assign and verify Azure RBAC roles at management group, subscription, resource group, and resource scope
- [ ] Build a management group hierarchy and understand policy inheritance across scopes
- [ ] Assign and evaluate Azure Policy definitions to enforce allowed locations
- [ ] Configure Cost Management budget alerts to control lab spending
- [ ] Run basic Azure CLI commands from LAB-W11-01 to list and inspect resources

MS Learn modules

- Manage Azure identities and governance

- Configure Azure Active Directory
- Configure user and group accounts
- Configure subscriptions and governance

Key Concepts

Term	Description
Entra ID	Microsoft's cloud identity platform (formerly Azure Active Directory); manages users, groups, and app registrations
Azure RBAC	Role-Based Access Control for Azure resources — controls *what* you can do with resources
Entra ID roles	Directory-level roles (e.g. Global Admin, User Admin) — control *who* can manage the directory itself, not resources
Management group	Hierarchical container above subscriptions; RBAC assignments and policies assigned here cascade to all child subscriptions
Azure Policy	Rules engine that evaluates resources against defined conditions and applies effects (Deny, Audit, Append, Modify, DeployIfNotExists)
Resource Group	Logical container for related Azure resources sharing a lifecycle; also a primary RBAC scope
Cost Management	Azure-native tool for monitoring spend, setting budgets, and triggering alerts when thresholds are reached
Resource lock	CanNotDelete or ReadOnly protection on resources; overrides RBAC — even Owners cannot delete a locked resource

Lab exercises (SSW-Lab + Azure portal)

Environment	Task
Azure portal	Create a *Resource Group*: `rg-sswlab-dev` in West Europe
Azure portal	Create an additional Entra ID user: `az-admin@<tenant>.onmicrosoft.com`
Azure portal	Assign the *Contributor* role to `az-admin` on the resource group via IAM
Azure portal	Create a *Management group* structure: Root → SSW → Dev/Prod
Azure portal	Assign an *Azure Policy*: "Allowed locations = West Europe, North Europe"
LAB-W11-01	Use Azure CLI: `az group list --output table`
Azure portal	Set up a *Cost budget alert* at €50 for `rg-sswlab-dev`

Lab commands

```
# Create a new Entra ID user
az ad user create --display-name "AZ Admin" --user-principal-name az-
admin@<tenant>.onmicrosoft.com --password "P@ssw0rd123!"

# Assign Contributor role on a resource group
az role assignment create --assignee az-admin@<tenant>.onmicrosoft.com --role
"Contributor" --scope /subscriptions/<sub-id>/resourceGroups/rg-sswlab-dev

# List all role assignments in a resource group
az role assignment list --resource-group rg-sswlab-dev --output table

# Assign a built-in Azure Policy (Allowed locations)
az policy assignment create --name "allowed-locations" --policy "e56962a6-4747-
49cd-b67b-bf8b01975c4f" --params '{"listOfAllowedLocations":{"value":
["westeurope","northeurope"]}}'

# Create a budget alert
az consumption budget create --budget-name "sswlab-budget" --amount 50 --time-
grain Monthly --resource-group rg-sswlab-dev --notifications
'[{"enabled":true,"operator":"GreaterThan","threshold":80,"contactEmails":
["admin@example.com"]}]'
```

Knowledge check

1. What is the difference between *Azure RBAC* and *Entra ID roles*?
2. How does *Policy* compare to *RBAC* — when do you use which?
3. What is the difference between *Management Group*, *Subscription*, *Resource Group* and *Resource*?
4. How does *Azure Cost Management* work and how do you configure budget alerts?

▶ Answers

Week 2 — Implement and manage storage

> **Exam domain:** Implement and manage storage · **Weight:** 15–20%

> **Real-world scenario:** A growing data analytics team at a retail client generates around 500 GB of raw data monthly. Reports older than 90 days are rarely accessed, and data older than two years must be kept for compliance but is virtually never read. The storage costs are rising fast and your

> Sogeti engagement lead asks you to design a lifecycle management strategy and recommend the right redundancy tier given that the team operates from a single Dutch office but the data is business-critical.

Learning Objectives

- [ ] Create and configure a General Purpose v2 storage account with LRS redundancy
- [ ] Understand and compare storage redundancy options: LRS, ZRS, GRS, RA-GRS, GZRS
- [ ] Generate and test a Shared Access Signature (SAS) token with scoped permissions
- [ ] Configure a lifecycle management policy to automatically tier blobs to Cool
- [ ] Create an Azure File Share and mount it via SMB from LAB-W11-01
- [ ] Install and configure Azure File Sync on LAB-DC01 with cloud tiering enabled

MS Learn modules

- Configure storage accounts
- Configure Azure Blob Storage
- Configure Azure Files and Azure File Sync
- Configure Azure Storage security

Key Concepts

Term	Description
LRS	Locally Redundant Storage — 3 copies within a single datacenter; no protection against datacenter failure
ZRS	Zone-Redundant Storage — copies across 3 availability zones in the same region
GRS	Geo-Redundant Storage — LRS in primary region + 3 copies in a secondary region (read-only via RA-GRS)
GZRS	Geo-Zone-Redundant Storage — combines ZRS in primary with geo-replication to secondary
Access tiers	Hot (frequent access), Cool (monthly), Cold (quarterly, 90-day min), Archive (rare, hours to rehydrate)
SAS token	Shared Access Signature — a time-limited, permission-scoped URL token; cannot be revoked unless tied to a Stored Access Policy
Stored Access Policy	A reusable, revocable access policy attached to a container; revoking it invalidates all associated SAS tokens
Azure File Sync	Syncs an on-premises Windows file server folder with an Azure File Share; supports cloud tiering
Cloud tiering	Files not accessed recently are evicted from local storage but remain accessible as stubs — fetched on demand from Azure

Term	Description
Lifecycle management	Rules that automatically move or delete blobs based on age or last-access patterns

Lab exercises (SSW-Lab + Azure portal)

Environment	Task
Azure portal	Create a Storage Account: LRS, General Purpose v2, Hot tier
Azure portal	Create a Blob container → upload a test file
Azure portal	Generate a *Shared Access Signature (SAS)* with read permissions, valid for 1 hour
LAB-W11-01	Use Azure Storage Explorer or `azcopy` to upload to blob
Azure portal	Create an *Azure File Share* → map via SMB (`net use Z: \\ ...`)
LAB-DC01	Install Azure File Sync agent → register server → sync an LAB-DC01 folder
Azure portal	Configure a *Lifecycle management policy*: move to Cool tier after 30 days

Lab commands

```
# Create a storage account
az storage account create --name sswlabstorage --resource-group rg-sswlab-dev --
location westeurope --sku Standard_LRS --kind StorageV2

# Generate a SAS token for a container (read-only, 1 hour)
az storage container generate-sas --account-name sswlabstorage --name mycontainer
--permissions r --expiry (Get-Date).AddHours(1).ToString("yyyy-MM-ddTHH:mmZ") --
output tsv

# Upload a file with azcopy
azcopy copy "C:\testfile.txt"
"https://sswlabstorage.blob.core.windows.net/mycontainer/testfile.txt?<SAS-
token>"

# List blobs in a container
az storage blob list --account-name sswlabstorage --container-name mycontainer --
output table

# Create an Azure File Share and mount it
az storage share create --name myfileshare --account-name sswlabstorage --quota 5
net use Z: \\sswlabstorage.file.core.windows.net\myfileshare
/u:AZURE\sswlabstorage <storage-key>
```

Knowledge check

1. What are the storage access tiers and when do you use each?
2. What is the difference between a *SAS token* and an *access key*?
3. How does *Azure File Sync* work and what is *cloud tiering*?
4. What is the difference between *LRS*, *ZRS*, *GRS* and *GZRS*?

▶ Answers

Week 3 — Deploy and manage Virtual Machines

> **Exam domain:** Deploy and manage Azure compute resources · **Weight:** 20–25%

> **Real-world scenario:** A logistics company is migrating its on-premises application tier to Azure. The application consists of two web servers and one application server that must remain available during scheduled platform maintenance windows. The client's SLA requires 99.95% uptime. You are the Azure administrator on the Sogeti project team and must design the VM deployment using Availability Sets, configure automated backups, and demonstrate a recovery procedure.

Learning Objectives

- ☐ Deploy a Windows Server 2022 VM using the Azure portal and configure its size, disk, and network settings
- ☐ Attach and initialise a managed data disk inside a running Windows VM
- ☐ Create a VM snapshot and restore a new managed disk from it
- ☐ Configure an Availability Set and understand fault domain vs update domain distribution
- ☐ Explain the differences between Availability Sets, Availability Zones, and VM Scale Sets
- ☐ Enable Azure Backup for a VM and execute an on-demand backup job

MS Learn modules

- Configure virtual machines
- Configure virtual machine availability
- Configure virtual machine extensions

Key Concepts

Term	Description
Availability Set	Groups VMs across Fault Domains (separate racks) and Update Domains (rolling reboot groups) within a single datacenter — protects against hardware failure, not datacenter failure

Term	Description
Availability Zone	Physically separate datacenters within a region; VMs in different zones are protected against datacenter-level failure
VM Scale Set (VMSS)	Auto-scaling group of identical VMs; scales out/in based on metrics (CPU, memory, custom); load-balanced automatically
Spot VM	Uses unused Azure capacity at a deeply discounted rate; can be evicted with 30 seconds' notice — suitable only for interruptible workloads
Managed disk	Azure-managed virtual disk; tiers: Premium SSD (production), Standard SSD (dev/test), Standard HDD (archive)
VM snapshot	Point-in-time read-only copy of a managed disk; used to create new disks or for rollback
Generalised image	VM image that has been sysprepped (Windows) or deprovision+generalised (Linux); all machine-specific state removed — suitable for deploying multiple identical VMs
Specialised image	Exact copy of a VM including machine identity and local accounts; used to restore a specific machine, not for bulk deployment
Azure Backup	Managed backup service using a Recovery Services Vault; supports VMs, Azure Files, SQL in VMs, and more

Lab exercises (Azure portal)

Environment	Task
Azure portal	Deploy a Windows Server 2022 VM (B2s SKU) in `rg-sswlab-dev`
Azure portal	Connect via RDP → install IIS via Server Manager
Azure portal	Add a *data disk* (32 GB, Standard SSD) → initialise in Windows
Azure portal	Take a *VM snapshot* → restore a new managed disk from the snapshot
Azure portal	Configure *Auto-shutdown* to control costs (22:00 UTC)
Azure portal	Deploy an *Availability Set* with 2 VMs → verify fault/update domain distribution
Azure portal	Enable *Azure Backup* for the VM → run an on-demand backup

Lab commands

```
# Deploy a VM via Azure CLI
az vm create --resource-group rg-sswlab-dev --name sswlab-vm01 --image
Win2022AzureEditionCore --size Standard_B2s --admin-username azureadmin --admin-
password "P@ssw0rd123!" --location westeurope

# Attach a data disk to an existing VM
az vm disk attach --resource-group rg-sswlab-dev --vm-name sswlab-vm01 --name
sswlab-datadisk01 --new --size-gb 32 --sku Standard_LRS

# Take a snapshot of the OS disk
$diskId = az vm show --resource-group rg-sswlab-dev --name sswlab-vm01 --query
"storageProfile.osDisk.managedDisk.id" -o tsv
az snapshot create --resource-group rg-sswlab-dev --name sswlab-snapshot01 --
source $diskId

# Enable Azure Backup for a VM
az backup protection enable-for-vm --resource-group rg-sswlab-dev --vault-name
sswlab-rsv --vm sswlab-vm01 --policy-name DefaultPolicy

# Trigger an on-demand backup job
az backup protection backup-now --resource-group rg-sswlab-dev --vault-name
sswlab-rsv --container-name sswlab-vm01 --item-name sswlab-vm01 --backup-
management-type AzureIaasVM
```

Knowledge check

1. What is the difference between *Availability Sets* and *Availability Zones*?
2. When do you use *Azure VM Scale Sets* versus standalone VMs?
3. How do *Azure Spot VMs* work and when are they appropriate?
4. What is the difference between *generalised* and *specialised* images?

▶ Answers

Week 4 — Containers and Azure App Service

> **Exam domain:** Deploy and manage Azure compute resources · **Weight:** 20–25%

> **Real-world scenario:** A software product team at a Sogeti client wants to modernise their release process. They currently deploy a monolithic .NET application to VMs manually, which causes 30-minute outages during each release. You are asked to migrate the application to Azure App Service,

> implement deployment slots for zero-downtime releases, and explore Azure Container Apps for a new microservices API that the team is building in parallel.

Learning Objectives

- [ ] Deploy an Azure App Service web app using Azure CLI from Cloud Shell with the F1 free tier
- [ ] Configure and execute a deployment slot swap for zero-downtime deployment
- [ ] Deploy an Azure Container Instance using both the portal and `az container create`
- [ ] Create an Azure Container Registry and push a container image to it
- [ ] Deploy an Azure Container App from ACR and configure horizontal scaling rules
- [ ] Explain the differences between ACI, ACA, and AKS and select the appropriate service

MS Learn modules

- Configure Azure App Service
- Configure Azure Container Instances
- Configure Azure Container Apps

Key Concepts

Term	Description
App Service Plan	Defines the compute tier (size and instance count) that backs one or more App Service apps; plans range from Free (F1) to Premium
Deployment slot	A separate instance of an App Service app (e.g. staging) in the same plan; supports slot swap for zero-downtime releases and instant rollback
Azure Container Registry (ACR)	Private Docker-compatible container registry hosted in Azure; integrates with ACI and ACA for private image pulls
Azure Container Instances (ACI)	Serverless container execution — starts in seconds, billed per second; no cluster required; ideal for short-lived or single-container workloads
Azure Container Apps (ACA)	Fully managed container platform built on Kubernetes (KEDA + Envoy); supports microservices, event-driven scaling, Dapr, and ingress — no direct Kubernetes management required
Azure Kubernetes Service (AKS)	Fully managed Kubernetes cluster; maximum control and flexibility but requires Kubernetes expertise to operate
Autoscaling	App Service and ACA both support horizontal scale-out based on CPU, memory, or HTTP request metrics; App Service plans must be Standard or higher for autoscale
Consumption plan	Serverless billing model for Azure Functions; scales to zero when idle — differs from an App Service Plan which has always-on compute

Lab exercises (Azure portal + Cloud Shell)

Environment	Task
Azure Cloud Shell	Deploy a simple web app: `az webapp create --sku F1 --name sswlab-app ...`
Azure portal	Configure a *deployment slot* (staging) → perform a *swap* to production
Azure portal	Deploy an Azure Container Instance with an nginx image
Azure Cloud Shell	`az container create --image nginx --dns-name-label sswlab-ci ...`
Azure portal	Create an *Azure Container Registry* → push a container image
Azure portal	Deploy a *Container App* from ACR → configure scaling rules
Azure portal	Configure an *App Service plan* → scale out to 2 instances
Azure portal	Review *App Service Diagnostics* → analyse availability graph

Lab commands

```
# Deploy an App Service web app (Free tier)
az webapp create --resource-group rg-sswlab-dev --plan sswlab-asp --name sswlab-app --runtime "DOTNET|8.0"

# Add a deployment slot
az webapp deployment slot create --name sswlab-app --resource-group rg-sswlab-dev --slot staging

# Swap staging to production
az webapp deployment slot swap --name sswlab-app --resource-group rg-sswlab-dev --slot staging --target-slot production

# Create a Container Registry and push an image
az acr create --resource-group rg-sswlab-dev --name sswlabacr --sku Basic
az acr login --name sswlabacr
docker tag myapp sswlabacr.azurecr.io/myapp:v1
docker push sswlabacr.azurecr.io/myapp:v1
```

Knowledge check

1. What is the difference between *Azure Container Instances*, *Azure Container Apps* and *Azure Kubernetes Service*?
2. How do *deployment slots* work and why is a *slot swap* useful?
3. What is the difference between an *App Service plan* and a *Consumption plan* for Azure Functions?
4. How do you configure *autoscaling* based on CPU usage?

▶ Answers

Week 5 — Virtual networks

> **Exam domain:** Implement and manage virtual networking · **Weight:** 15–20%

> **Real-world scenario:** A healthcare client is deploying a patient data processing application in Azure. Regulatory requirements state that no patient data may traverse the public internet, and the storage account and database endpoints must only be reachable from within the corporate Azure VNet. The client's on-premises network must also connect securely to the Azure environment. You are responsible for designing the network segmentation, configuring Private Endpoints, and setting up the VPN connection from the Sogeti-managed lab environment.

Learning Objectives

- ☐ Create a VNet with multiple subnets and assign appropriate address spaces
- ☐ Create and configure NSG rules, and understand the difference between subnet-level and NIC-level application
- ☐ Configure VNet peering between two VNets and verify connectivity
- ☐ Create an Azure Private DNS Zone and enable automatic VM registration
- ☐ Configure a Point-to-Site VPN from LAB-W11-01 to an Azure VNet
- ☐ Deploy Azure Bastion and connect to a VM without a public IP or open RDP port
- ☐ Create a Private Endpoint for a Storage Account and validate DNS resolution

MS Learn modules

- Configure virtual networks
- Configure network security groups
- Configure Azure DNS
- Configure virtual network peering

Key Concepts

Term	Description
VNet	Virtual Network — an isolated layer-3 network in Azure; resources within a VNet communicate by default, resources in different VNets do not
NSG	Network Security Group — a stateful packet filter with inbound and outbound rules based on IP, port, and protocol; applied to a subnet or a NIC
Service tag	A named group of IP prefixes for an Azure service (e.g. `AzureLoadBalancer`, `Internet`, `Storage`) used in NSG rules to avoid hardcoding IP ranges

Term	Description
VNet peering	Direct, low-latency connection between two VNets over the Azure backbone; non-transitive by default
Private DNS Zone	DNS resolution scoped to linked VNets; names resolve to private IP addresses and are never exposed to the internet
Azure Bastion	Managed PaaS RDP/SSH gateway deployed into a dedicated `AzureBastionSubnet` (/26 or larger); provides secure browser-based access without exposing VM ports publicly
Service endpoint	Secures a subnet's outbound route to an Azure PaaS service via the Azure backbone — the PaaS service still has a public IP
Private endpoint	Places an Azure PaaS service (storage, SQL, etc.) behind a private IP in your VNet; traffic never traverses the public internet
Point-to-Site VPN	VPN connection from an individual device (e.g. LAB-W11-01) to an Azure VNet; uses certificate or Entra ID authentication

Lab exercises (Azure portal + SSW-Lab)

Environment	Task
Azure portal	Create a VNet: `10.100.0.0/16` with subnets `frontend/10.100.1.0/24` and `backend/10.100.2.0/24`
Azure portal	Configure an *NSG*: allow HTTP/HTTPS inbound, deny everything else
Azure portal	Attach NSG to the frontend subnet → test from a VM in the backend subnet
Azure portal	Create a second VNet → configure *VNet peering* between both
Azure portal	Configure an *Azure Private DNS Zone* → auto-register VMs
LAB-W11-01	Configure a *Point-to-Site VPN* to the Azure VNet → test the connection
Azure portal	Deploy *Azure Bastion* → connect to a VM without public IP or RDP port
Azure portal	Configure a *Private Endpoint* for a Storage Account → verify DNS resolution
Azure portal	Review *Effective routes* on the NIC of a VM

Lab commands

```
# Create a VNet with two subnets
az network vnet create --resource-group rg-sswlab-dev --name sswlab-vnet --
address-prefix 10.100.0.0/16 --subnet-name frontend --subnet-prefix 10.100.1.0/24
az network vnet subnet create --resource-group rg-sswlab-dev --vnet-name sswlab-
vnet --name backend --address-prefix 10.100.2.0/24

# Create an NSG and attach to a subnet
az network nsg create --resource-group rg-sswlab-dev --name sswlab-nsg-frontend
az network nsg rule create --resource-group rg-sswlab-dev --nsg-name sswlab-nsg-
frontend --name AllowHTTP --priority 100 --protocol Tcp --destination-port-range
80 443 --access Allow --direction Inbound
az network vnet subnet update --resource-group rg-sswlab-dev --vnet-name sswlab-
vnet --name frontend --network-security-group sswlab-nsg-frontend

# Configure VNet peering (both directions)
az network vnet peering create --resource-group rg-sswlab-dev --name vnet1-to-
vnet2 --vnet-name sswlab-vnet --remote-vnet sswlab-vnet2 --allow-vnet-access
az network vnet peering create --resource-group rg-sswlab-dev --name vnet2-to-
vnet1 --vnet-name sswlab-vnet2 --remote-vnet sswlab-vnet --allow-vnet-access
```

Knowledge check

1. What is the difference between an *NSG* applied at the subnet level and at the NIC level?
2. How does *VNet peering* work — is it transitive?
3. What is the difference between *Azure DNS* public zones and *Private DNS Zones*?
4. What is a *service endpoint* versus a *private endpoint*?

▶ Answers

Week 6 — Load balancing and network routing

> **Exam domain:** Implement and manage virtual networking · **Weight:** 15–20%

> **Real-world scenario:** A Sogeti client runs a public-facing e-commerce platform with a web tier and an API tier, both hosted on Azure VMs behind a load balancer. During a recent peak sales event, one web server became unresponsive but continued receiving traffic, causing errors for customers. You are asked to implement proper health probing, configure path-based routing to separate web and API traffic, and use Azure Network Watcher to diagnose an intermittent connectivity issue reported by the operations team.

Learning Objectives

- [ ] Deploy a Standard Load Balancer with a backend pool, health probe, and load balancing rule
- [ ] Verify Load Balancer failover by stopping one backend VM and observing traffic redistribution
- [ ] Create User Defined Routes (UDR) to redirect traffic via a network virtual appliance
- [ ] Deploy an Application Gateway with WAF and configure path-based routing rules
- [ ] Use Azure Network Watcher IP Flow Verify and Connection Troubleshoot to diagnose connectivity issues
- [ ] Explain the differences between Azure Load Balancer, Application Gateway, and Azure Front Door

MS Learn modules

- Configure Azure Load Balancer
- Configure Azure Application Gateway
- Configure network routing and endpoints
- Configure Azure Firewall

Key Concepts

Term	Description
Azure Load Balancer (L4)	Distributes TCP/UDP traffic based on IP and port; no content inspection; supports Standard (zone-redundant) and Basic SKU
Application Gateway (L7)	HTTP/HTTPS load balancer with URL path routing, SSL termination, session affinity, header rewriting, and integrated WAF
Azure Front Door	Global HTTP/HTTPS load balancer and CDN with routing across regions; used for geo-redundant multi-region apps
WAF (Web Application Firewall)	Protects against OWASP Top 10 threats; available on Application Gateway and Front Door
Health probe	Periodically tests backend VMs; if a probe fails, the Load Balancer stops sending traffic to that instance
UDR (User Defined Route)	A custom route table entry that overrides Azure's default system routes; commonly used to force traffic through a firewall/NVA
SNAT	Source Network Address Translation — a Load Balancer uses SNAT to replace the source IP of outbound traffic from backend VMs with the Load Balancer's frontend IP
NVA (Network Virtual Appliance)	A VM running firewall or routing software (e.g. Palo Alto, Fortinet); traffic is steered to it via UDR
Network Watcher	Suite of network diagnostic tools: IP Flow Verify, Connection Troubleshoot, NSG Flow Logs, Packet Capture, Next Hop

Lab exercises (Azure portal)

Environment	Task
Azure portal	Deploy a *Standard Load Balancer* for 2 web server VMs
Azure portal	Configure a health probe on port 80 → test failover (stop 1 VM)
Azure portal	Configure *User Defined Routes (UDR)*: route traffic via a Network Virtual Appliance
Azure portal	Deploy an *Application Gateway* with WAF → configure a path-based routing rule
Azure portal	Use *Azure Network Watcher* → *IP Flow Verify* to diagnose NSG blocking
Azure portal	Use *Connection Troubleshoot* to analyse connectivity issues

Lab commands

```
# Create a Standard Load Balancer with a public IP
az network lb create --resource-group rg-sswlab-dev --name sswlab-lb --sku
Standard --frontend-ip-name frontendConfig --public-ip-address sswlab-lb-pip

# Add a health probe
az network lb probe create --resource-group rg-sswlab-dev --lb-name sswlab-lb --
name httpProbe --protocol Http --port 80 --path /

# Add a load balancing rule
az network lb rule create --resource-group rg-sswlab-dev --lb-name sswlab-lb --
name httpRule --protocol Tcp --frontend-port 80 --backend-port 80 --frontend-ip-
name frontendConfig --backend-pool-name backendPool --probe-name httpProbe

# Use Network Watcher IP Flow Verify
az network watcher test-ip-flow --resource-group rg-sswlab-dev --vm sswlab-vm01 -
-direction Inbound --protocol Tcp --local 10.100.1.4:80 --remote
203.0.113.10:12345
```

Knowledge check

1. What is the difference between *Azure Load Balancer* (L4) and *Application Gateway* (L7)?
2. When do you use *Azure Front Door* versus Application Gateway?
3. How does *SNAT* work in a Load Balancer configuration?
4. What are *service tags* in NSG rules and what are they used for?

▶ Answers

Week 7 — Monitoring and Azure Monitor

> **Exam domain:** Monitor and maintain Azure resources · **Weight:** 10–15%

> **Real-world scenario:** A Sogeti client running a production ERP system on Azure VMs reports intermittent performance degradation on Monday mornings. The operations team has no centralised logging and relies on ad-hoc RDP sessions to check Event Viewer. You are asked to implement Azure Monitor with Log Analytics, configure automated CPU alerts, and set up Azure Backup with a proven restore procedure — all before the client's next quarterly review.

Learning Objectives

- [ ] Create a Log Analytics Workspace and connect Azure VMs to it
- [ ] Enable VM Insights and interpret the Performance and Map tabs
- [ ] Write KQL queries to filter, aggregate, and project log data
- [ ] Create an alert rule with a metric trigger and configure an action group for email notification
- [ ] Install the Azure Monitor Agent (AMA) on LAB-DC01 and verify data ingestion
- [ ] Configure a Recovery Services Vault, back up DC01 files, and perform a test file restore
- [ ] Configure Azure Site Recovery for a VM and execute a test failover to a secondary region

MS Learn modules

- Configure Azure Monitor
- Configure Log Analytics
- Configure Azure alerts and action groups
- Configure Azure Backup and recovery

Key Concepts

Term	Description
Azure Monitor Metrics	Numerical time-series data (CPU %, disk I/O, network bytes) collected every minute; retained 93 days; best for real-time dashboards and metric-based alerts
Azure Monitor Logs	Text-structured data stored in a Log Analytics workspace; queried with KQL; retained 30 days by default (configurable up to 2 years)
KQL	Kusto Query Language — used to query Log Analytics data; key operators: `where`, `project`, `summarize`, `order by`, `join`, `render`
Azure Monitor Agent (AMA)	Current unified agent for collecting logs and metrics from VMs (replaces MMA/OMS agent); deployed via Data Collection Rules
Action group	A reusable collection of notification and action targets (email, SMS, webhook, Azure Function, ITSM) triggered by alert rules

Term	Description
Alert rule	Monitors a metric or log query result; fires when a condition is breached; triggers an action group
Recovery Services Vault	Container for Azure Backup and Azure Site Recovery configurations and recovery points
Azure Backup	Protection against data loss — stores recovery points; supports VMs, Azure Files, SQL Server in VMs, blobs
Azure Site Recovery (ASR)	Disaster recovery — continuously replicates VM disk changes to a secondary region; enables failover if the primary region fails
Diagnostic settings	Per-resource configuration that routes platform logs and metrics to a Log Analytics workspace, storage account, or Event Hub

Lab exercises (Azure portal + SSW-Lab)

Environment	Task
Azure portal	Create a *Log Analytics Workspace* → connect the Azure VMs
Azure portal	Enable *VM Insights* → review the Performance and Map tabs
Azure portal	Write a KQL query: CPU > 80% in the past 24 hours
Azure portal	Create an *Alert rule*: CPU > 85% → email notification to admin
LAB-DC01	Install the *Azure Monitor Agent (AMA)* → verify in Log Analytics
Azure portal	Configure a *Recovery Services Vault* → back up DC01 files
Azure portal	Run a *test restore* → recover a file to an alternate location
Azure portal	Configure *Azure Site Recovery* for a VM → perform a test failover to a secondary region

Lab commands

```
# Query CPU usage above 80% for a VM in the last 24 hours (KQL in Log Analytics)
Perf
| where TimeGenerated > ago(24h)
| where CounterName == "% Processor Time"
| where CounterValue > 80
| project TimeGenerated, Computer, CounterValue
| order by TimeGenerated desc

# Create a Log Analytics workspace via CLI
az monitor log-analytics workspace create --resource-group rg-sswlab-dev --
workspace-name sswlab-law --location westeurope

# Create an action group for email alerts
az monitor action-group create --resource-group rg-sswlab-dev --name sswlab-ag --
short-name ssw-ag --email-receiver name=AdminEmail email=admin@example.com

# Create a CPU metric alert rule
az monitor metrics alert create --resource-group rg-sswlab-dev --name "HighCPU-
Alert" --scopes /subscriptions/<sub-id>/resourceGroups/rg-sswlab-
dev/providers/Microsoft.Compute/virtualMachines/sswlab-vm01 --condition "avg
Percentage CPU > 85" --window-size 5m --evaluation-frequency 1m --action sswlab-
ag
```

Knowledge check

1. What is the difference between *Azure Monitor Metrics* and *Azure Monitor Logs*?
2. How does *Azure Alerts* work — what are *action groups*?
3. Write a KQL query: all failed sign-ins from the past week.
4. What is the difference between *Azure Backup* and *Azure Site Recovery*?

▶ Answers

Week 8 — Exam preparation

Activities

- Review weak domains based on the official exam study guide
- Complete the **Microsoft Learn practice assessment** for AZ-104: Practice assessment
- Practice resource creation via **Azure CLI** and **PowerShell (Az module)** — the exam includes CLI questions
- Revisit VNet architecture and NSG scenarios (most frequently examined)
- Clean up all Azure resources to avoid charges: `az group delete -n rg-sswlab-dev`

- Schedule your exam via Pearson VUE

Exam focus areas

- Networking: NSG rules, UDR, VNet peering, Private Endpoints, Azure Bastion — heavily weighted
- VM availability: know the exact difference between Availability Set / Zone / VMSS
- Storage: redundancy options (LRS/ZRS/GRS), tier management, blob versioning, soft delete
- Compute: ARM templates *and* Bicep files — both are in scope
- RBAC: scope levels (management group → subscription → resource group → resource)
- Backup & recovery: difference between Azure Backup and Azure Site Recovery (failover)
- KQL: basic queries (where, summarize, project) appear in case-study questions

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Exam Coverage Gaps and Must-Do Labs

This section maps the SSW-Lab Minimal preset (DC01 + W11-01 as hybrid endpoints; all Azure work in the cloud) against the full AZ-104 exam domain coverage. It identifies topics that are underrepresented in the 8-week programme and prescribes the additional labs most likely to close scoring gaps.

Topics not fully covered by the weekly labs

Gap area	Why it matters	Suggested action
ARM templates and Bicep	Exam includes hands-on deployment scenarios with both formats	Complete Labs 4.7–4.8 from the M365-Lab reference guide; export a resource group as ARM JSON, modify a parameter, redeploy; author a minimal Bicep file and deploy via `az deployment group create`
VM Scale Sets with autoscale	Heavily tested in compute domain (20–25%); VMSS scaling scenarios appear in case studies	Deploy a VMSS with 2 instances; configure CPU-based scale-out (>80%) and scale-in (<20%) rules; verify instance count changes
Storage: soft delete, versioning, object replication	Exam asks detailed questions about blob data protection options	Enable soft delete (7-day retention) and blob versioning on a storage account; delete and restore a blob; configure object replication between two accounts
Storage firewall and private endpoint combined	Common exam scenario: restrict storage access to a specific VNet	Configure storage firewall to deny public access; add a subnet service endpoint; then replace with a private endpoint and verify DNS resolves to private IP
Azure Disk Encryption	Part of compute domain; appears in security-oriented questions	Enable ADE on a VM using an Azure Key Vault; verify encryption status with `Get-AzVmDiskEncryptionStatus`
Backup vault (DataProtection API)	Exam distinguishes Recovery Services vault from Backup vault	Create a Backup vault; configure a blob backup policy; understand which workloads require which vault type

Gap area	Why it matters	Suggested action
NSG flow logs and Traffic Analytics	Network Watcher tools are in-scope for the monitoring domain	Enable NSG flow logs on a subnet NSG; send to a storage account and Log Analytics; query `NTANetAnalytics` in KQL
ExpressRoute vs VPN Gateway decision	Appears in architecture scenario questions	Study conceptually (ExpressRoute cannot be lab-tested due to cost); focus on: when private connectivity is required, latency vs cost trade-offs, SKU capabilities

Must-do labs before scheduling the exam

1. **ARM + Bicep deployment**: Export `rg-sswlab-dev` as an ARM template → edit a parameter value → redeploy in Incremental mode. Then write a minimal Bicep file to create a storage account and deploy it from Azure CLI or Cloud Shell.
2. **VM Scale Set with autoscale**: Create a VMSS (Windows Server 2022, 2 initial instances) in `rg-sswlab-dev` → configure a CPU-based autoscale profile (min 1 / max 5) → generate CPU load and observe scale-out → verify in Metrics.
3. **Storage data protection scenario**: On one storage account, enable soft delete (7 days), blob versioning, and a lifecycle policy (Cool after 30 days, Archive after 90 days) → delete a blob and restore it → inspect version history.
4. **Storage access restriction**: Configure the storage account firewall to Deny public access → add a VNet subnet rule → validate access is blocked from an unlisted source → replace with a private endpoint and confirm DNS resolution returns a private IP.
5. **Azure Site Recovery test failover**: In the Recovery Services Vault, enable replication for one Azure VM to North Europe → wait for Protected status → run a Test Failover → document the failover RTO → clean up the test environment and verify Failback capability.

Exit criteria before booking the exam

1. You can demonstrate all five AZ-104 exam domains hands-on via the Azure portal, Azure CLI, and PowerShell — without consulting documentation.
2. You can diagnose a network connectivity failure end-to-end using Network Watcher (IP Flow Verify, Effective routes, NSG Flow Logs).
3. You score 80%+ consistently on the Microsoft Learn practice assessment.
4. You have completed at least two full review passes over identity/governance, networking, and backup/recovery — the three domains most frequently cited as exam weak points.